

Zadanie 7

Rozwiąż równanie $\operatorname{tg}^4 x - 4\operatorname{tg}^2 x + 3 = 0$.

① Podstawiamy zmienną pomocniczą.

$$t = \operatorname{tg}^2 x \quad t \in [0, +\infty) \quad , \quad x \neq \frac{\pi}{2} + k\pi$$

$$t^2 - 4t + 3 = 0$$

$$\Delta = 16 - 12 = 4$$

$$t_1 = \frac{4-2}{2} = 1$$

$$t_2 = \frac{4+2}{2} = 3$$

$$(t-1)(t-3) = 0$$

$$t = 1 \quad t = 3$$

② Łaemy podstawienie.

$$\operatorname{tg}^2 x = 1 \quad \operatorname{tg}^2 x = 3$$

$$\begin{array}{ccccccc} \operatorname{tg} x = 1 & \vee & \operatorname{tg} x = -1 & \vee & \operatorname{tg} x = \sqrt{3} & \vee & \operatorname{tg} x = -\sqrt{3} \\ x = \frac{\pi}{4} + k\pi & & x = -\frac{\pi}{4} + k\pi & & x = \frac{\pi}{3} + k\pi & & x = -\frac{\pi}{3} + k\pi \quad , \quad k \in \mathbb{Z} \end{array}$$

Odpo.: $x = \frac{\pi}{4} + k\pi$, $x = -\frac{\pi}{4} + k\pi$, $x = \frac{\pi}{3} + k\pi$, $x = -\frac{\pi}{3} + k\pi$, $k \in \mathbb{Z}$.